Major Advanced Topics

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Magnetic-Field Effect in High-Order Above-Threshold Ionization

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Magnetic-Field Effect as a Tool to Investigate Electron Correlation in Strong-Field Ionization (Phys. Rev. Lett. 128, 113201 (2022))

Surrounding concepts

- Electron correlation

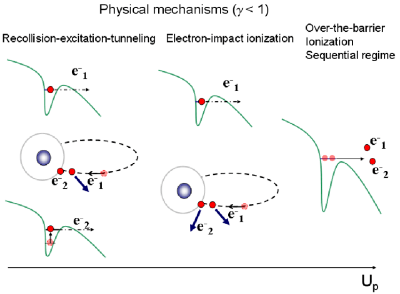
$$\Psi(\boldsymbol{r}_1,\boldsymbol{r}_2)\Theta(1,2) = \frac{1}{2}\left[\varphi_a(\boldsymbol{r}_1)\varphi_b(\boldsymbol{r}_2) \pm \varphi_b(\boldsymbol{r}_1)\varphi_a(\boldsymbol{r}_2)\right] \times \left[\sigma_a(1)\sigma_b(2) \pm \sigma_b(1)\sigma_a(2)\right]$$

$$\Psi_H(\boldsymbol{r}_1,\boldsymbol{r}_2)\Theta(1,2) = \varphi_a(\boldsymbol{r}_1)\varphi_b(\boldsymbol{r}_2)\sigma_a(1)\sigma_b(2)$$

- Sequential Double Ionization
- Nonsequential Double Ionization (NSDI)
  - Successfully explained in the dipole approx. for a wide range of intensities and wavelengths
  - Can happen a number of different ways

$$U_p = \frac{F_0^2}{4\omega^2}, \qquad \gamma = \sqrt{\frac{I_p}{2U_p}}$$

- 3 steps model



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The state of the problem

- In the nondipole case
- Conservation of momentum for a single electron ionization

$$\langle p_{x,e}\rangle (E_e) = \frac{E_e}{c} + \frac{1}{3}\frac{I_p}{c}.$$

- Forward shift of $I_p/(3c)$ caused by ionization in itself and tunneling through the barrier
- What about double ionization?
  - The naive assumption

$$\langle p_{x1} + p_{x2}\rangle (E_{\text{sum}}) = \frac{E_{\text{sum}}}{c} + \frac{1}{3} \times \frac{I_{p1} + I_{p2}}{c}.$$

- Yes, but what about correlated electrons, as in NDSI?
  - Emmanouilidou *et al* think that the sum nondipole shift will be enhanced
  - Sun *et al* recently confirmed

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What to expect from this paper

- The magnetic field is used to investigate the electron correlation in NDSI of Xe atoms
- Their finding is twofold
  - Observed nondipole shift coincide with classical prediction for free electrons

$$\langle p_{x1} + p_{x2} \rangle \left(E_{\mathrm{sum}} \right) = \frac{E_{\mathrm{sum}}}{c}.$$

  - Observed nondipole shifts for emitted pairs in same and opposite directions are congruent
- Here, returning e^- is recaptured by the ion in an excited state, exciting the other. Subsequent double tunnel ionization follows.
- That explanation of NDSI through an intermediate double excited state is supported by the facts that
  - I'_{p1} and I'_{p2} close to 0, so that $(I'_{p1} + I'_{p2})/(3c) \longrightarrow 0$
  - Recollisional nondipole shift is transfered to the ion

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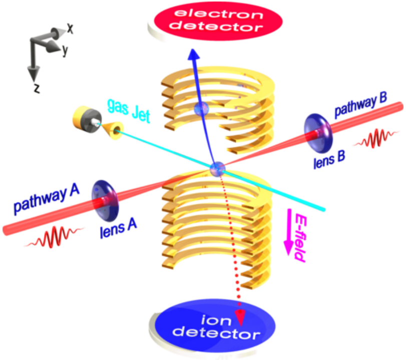
Experimental setup

- Making use of the following strategy
  - (800 nm, 10 kHz, 25 fs), Coherent Legend Elite Duo
  - Split, then focused back from 2 opposite directions on the same spot of supersonic Xe gas.
  - Symmetry of the experiment along the polarization direction (z-axis) prompted the symmetrisation of the data in regard to that same direction.
- Ratio Xe^{2+}/Xe^{+} is compared to Chaloupka *et al*
- Static E-field is applied to guide e^- and ions to opposite directions with a static field of $29.8\,\mathrm{V/cm}$ to times- and position-sensitive detectors
- Momenta of the e^- are retrieved from time of flight and positions of impact



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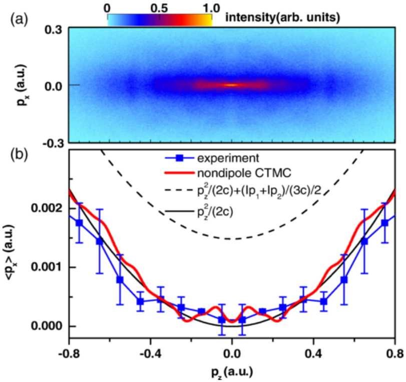
Electron momentum distribution measured along the $z-$ and $x-$ axis

- Detection of single event is $0.03\,\mathrm{a.u.}$ in p_x-, p_y- and $0.3\,\mathrm{a.u}$ in p_z-direction
- Coincidentally measured with Xe^{2+}
- First and second e^- are both taken into account and undistinguishable
- Fig. 1(a) shows the $2\mathrm{D}$ electron momentum the axis
    - Measured with the help of a COLTRIMS microscope whose resolution is given above
- Fig. 1(b) shows the mean electron momentum along the $z-$axis
    - Strikingly coincides with classical prediction for a free electron
    $$\langle p_x \rangle = p_z^2/2c$$
    - No under-barrier enhancement $(I_{p1} + I_{p2})/(3c)$ is observed



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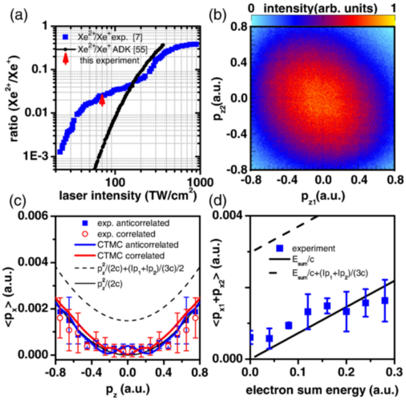
The electron-correlation detail

- Fig. 2(a) Ratio $\mathrm{Xe}^{2+}/\mathrm{Xe}^{+}$
  - Result of the experiment located in the knee-structure region
  - Process is dominated by NSDI
- Fig. 2(b) Measured electron-electron correlation map
  - Agreement with X. Sun *et al.*, Phys. Rev. Lett. 113 (2014)
  - Events in the 1st and 3rd quadrants correspond to cases where both e^- are ejected in the same hemisphere (correlated emissions)
- Fig. 2(c) Correlated and anticorrelated case one electron momentum
  - Correlated and anticorrelated curves are congruent
  - Contrast with results from F. Sun *et al*, Phys. Rev. A 101 (2020) with Ar
- Fig. 2(d) mean sum momentum
  - Selection of events 2 e^- and Xe^{2+} measured
  - Further shows that the classical prediction is followed



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The underlying phyics

- Tunneling process is described in the ADK model

- Magnetic field is "artificially" taken into account in the initial transverse direction of

$$W(v_x, v_y) = \frac{2\sqrt{2I_{p1}}}{|F(\tau)|} \times \exp\left\{\left[\left(v_x - \frac{I_{p1}}{(3c)}\right)^2 + v_y^2\right]\frac{\sqrt{2I_{p1}}}{|F(\tau)|}\right\},$$

$I_{p1} = 0.45\,\mathrm{a.u.}$ and $\tau = t - x/c$

- After tunneling, the subsequent evolution is modeled using Newton

$$\frac{d^2\boldsymbol{r}}{dt^2} = -\boldsymbol{\nabla}_{\boldsymbol{r}}\left(V_{ne}^{\mathrm{GSZ}} + V_{ee}\right) + q\boldsymbol{F}(\tau) + q\boldsymbol{v}\times\boldsymbol{B}(\tau), \quad Z = 54$$

$$V_{ne}^{\mathrm{GSZ}}(r) = -\left[2 + (Z-2)\Omega(r)\right]/r, \quad \Omega(r) = 1/\left[(\eta/\xi)(e^{\xi r - 1} + 1)\right], \quad \eta = 1.1701, \quad \xi = 5.2075,$$

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$$\boldsymbol{F}(\tau) = F_0 f(\tau)\cos(\omega\tau)\hat{\boldsymbol{e}}_z, \qquad \boldsymbol{B}(\tau) = \hat{\boldsymbol{e}}_x \times \boldsymbol{F}(\tau)/c, \qquad I_{p2} = 0.78\,\mathrm{a.u.},$$

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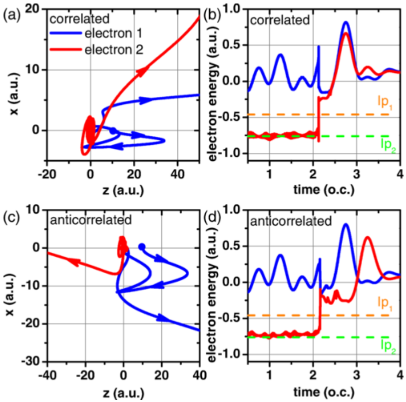
Dynamics of the NDSI

- Both cases
  - First electron can be seen cricling the ionic core
  - Rescattering, recombination and formation of Rydberg states can be seen through the energy diagrams
  - Both electrons are ejected from some excited sates, which explains why the enhancement term is absent
  - In the case of the first electron, enhancement $(I_{p1})/(3c)$ gained through tunneling is transferred to the ion at recombination
- Correlated case Figs. 3(a) and (b)
  - Both electrons ionized within the same half cycle
  - They go the same direction
- Anticorrelated case Figs. 3(c) and (b)
  - Both electrons ionized within the same half cycle
  - They go opposite directions



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Closure

- The electron correlation in the case of Xe has been studied
- There is no enhancement term for neither of the emitted electrons during the ionization
- Correlated and anticorrelated electrons have congruent energies
- There is a strike contrast between the results obtained for low-Z atoms like Ar

- Those findings pave the way to further investigating the nondipole effect in strong-field driven electron correlation dynamics